

Would an Iranian Closure of the Strait of Hormuz Trigger a Worldwide Recession?

Executive summary

A **full, sustained closure** of the Strait of Hormuz would constitute one of the largest plausible energy-supply disruptions in modern history because the Strait is the outlet for **~20 million barrels per day (mb/d)** of crude oil, condensate, and refined products in recent years and the single biggest maritime chokepoint for oil by volume. In 2025 specifically, the International Energy Agency (IEA) estimates **19.87 mb/d** of oil exports (14.95 mb/d crude+condensates and 4.93 mb/d products) transited the Strait. ¹

Even if Saudi Arabia and the UAE maximized bypass pipelines, **only a minority of those flows could be rerouted**. The IEA estimates **3.5–5.5 mb/d** of available alternative export capacity (Saudi East-West/Petroline and the UAE's Abu Dhabi Crude Oil Pipeline to Fujairah), and it stresses that logistics to sustain rerouting at the top end are not fully proven. ² EIA analysis is even more conservative for “effective unused” bypass capacity (**≈2.6 mb/d**) in the event of disruption. ³ A **complete closure** therefore implies an **immediate physical export shortfall** on the order of **~14–17 mb/d** after the initial ramp-up of bypass routes (central estimate ~15 mb/d), with additional severe refined-product dislocations (because most bypass capacity is crude-focused). ⁴

A closure would also hit natural gas through LNG. The Strait is the route for **~20% of global LNG trade**; IEA estimates **just over 112 bcm** transited Hormuz in 2025 (≈almost 20% of global LNG trade) and notes there are **no alternative routes** to bring most Qatari and UAE LNG to market. ⁵ This makes a Hormuz closure a **dual oil+LNG shock**, raising the odds of a synchronized global downturn.

Whether a “worldwide recession” occurs depends heavily on **duration** and on the degree to which inventories, emergency stock releases, demand destruction, and non-Gulf supply response can bridge the gap. Using IMF research cited by the IMF's Chief Economist—a **10% oil price increase reduces global output ~0.15% and raises global inflation ~0.4 percentage point in the following year**—a sustained doubling or tripling of oil prices (plausible under a multi-month closure given the scale of the physical shortfall) would push the global economy toward recessionary territory, especially when combined with LNG-driven power and industrial curtailments. ⁶

A calibrated judgment based on the above data and historical analogs is that: - **Scenario A (complete closure for 1 month)**: global recession risk is **material but not dominant** (illustrative probability **~15–30%**), primarily because markets may expect reopening and because emergency measures can bridge short gaps—though at significant inflation and confidence cost. ⁷

- **Scenario B (3 months)**: recession risk becomes **high** (illustrative probability **~35–55%**) as inventories and emergency releases become harder to sustain and both oil and LNG markets must clear via price-driven demand destruction. ⁸

- **Scenario C (6 months)**: recession risk is **very high** (illustrative probability **~55–75%**) because the shock becomes large enough and long enough to force widespread rationing/curtailments in the most-exposed

Asian economies and to trigger tighter monetary/financial conditions globally. ⁹

- **Scenario D (indefinite until rerouting saturates and demand rebalances):** recession risk is **extreme** (illustrative probability **~75–90%**), because the world would be forced into a prolonged regime of high energy prices, LNG scarcity, and persistent trade/income losses for importers. ¹⁰

These probabilities are not “statistical forecasts”; they are structured judgments grounded in observed volumes, rerouting constraints, and published macro multipliers, with large uncertainty bands.

Baseline traffic, volumes, and exposure

Recent oil and LNG volumes through Hormuz show the chokepoint remains structurally critical. EIA chokepoint statistics imply total oil flows through Hormuz were roughly **~19–22 mb/d** over 2020–2024, with **LNG ~10–11+ Bcf/d**. ¹¹ EIA also emphasizes that flows through Hormuz in 2024 and 1Q25 were **more than one-quarter of global seaborne oil trade** and about **one-fifth of global oil and petroleum product consumption**, with **~one-fifth of global LNG trade** also transiting in 2024. ¹²

Recent volumes through the Strait of Hormuz

Metric	2021	2022	2023	2024	1H25
Total oil flows (mb/d)	19.7	21.9	21.8	20.7	20.9
Crude & condensate (mb/d)	14.7	16.2	15.8	14.6	14.7
Petroleum products (mb/d)	5.1	5.6	6.0	6.1	6.1
LNG (Bcf/d)	10.8	11.0	10.6	10.5	11.4

Data: EIA “World Oil Transit Chokepoints” table for Hormuz. ¹¹

Total shipping traffic and tanker transits

AIS-based experimental statistics from the UK Office for National Statistics (ONS) Data Science Campus show that **weekly crossings through the Strait of Hormuz** were typically several hundred per week in 2023 (visualized across passages). ¹³ Digitizing the ONS chart for 2023 implies an average of **~613 trading-ship crossings/week**, or roughly **~31,900 crossings/year** (order-of-magnitude ~30k+ annual crossings). ¹³

Digitizing the ONS “by ship type” series implies (2023): - **Oil tanker crossings: ~197/week, ≈ 10,300/year**. ¹⁴

- **Liquefied gas tanker crossings: ~84/week, ≈ 4,400/year**. ¹⁴

Important caveat: ONS ship-type bins are “oil” and “liquefied gas” tankers (and others); **liquefied gas includes LNG and other liquefied gases**, so it is not a pure LNG-carrier count. ¹⁵

For tonnage-based “total shipping” intensity, UNCTAD (using Clarksons Shipping Intelligence Network) shows **monthly ship transits through the Strait of Hormuz in the ~180–210 million gross tons range** in

the period **Sep 2023–Jun 2025**, with the chart showing **~192 million gross tons** in the latest month plotted.

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Major exporters and major importers

The IEA provides a detailed 2025 breakdown (Kpler-based) of oil exports transiting Hormuz by **exporter** and **destination**:

Exports by origin via Hormuz (2025, mb/d)

Saudi Arabia is the largest exporter transiting Hormuz (**≈6.23 mb/d** total), followed by Iraq (**≈3.63**), UAE (**≈3.24**), Iran (**≈2.41**), Kuwait (**≈2.37**), and Qatar (**≈1.43**). Total **≈19.87 mb/d**. ¹

Crude export destinations via Hormuz (2025, mb/d)

China **≈4.6**, India **≈2.1**, Other Asia **≈6.2**, Europe **≈0.6**, Americas **≈0.5**, Africa **≈0.2**. ¹⁷

Oil product export destinations via Hormuz (2025, mb/d)

China **≈0.8**, India **≈0.7**, Other Asia **≈2.1**, Europe **≈0.5**, Americas **≈0.2**, Africa **≈0.6**. ¹⁷

The IEA also notes that **China and India combined received 44%** of crude exports passing through the Strait in 2025, and only **~600 kb/d (~4%)** of crude flows were routed into Europe. ¹⁸ Consistent with this, EIA estimates that **84% of crude/condensate** and **83% of LNG** moving through Hormuz in 2024 went to Asian markets, and that China, India, Japan, and South Korea together accounted for **69%** of Hormuz crude/condensate flows in 2024. ¹⁹

Disruption scenarios and physical supply impacts

This section models **complete closure** scenarios under the user's requested A–D structure. The key mechanical fact is that a closure prevents exports from most Persian Gulf terminals from reaching open ocean, so the relevant question becomes: **how much can bypass pipelines and non-Hormuz routes replace, and how fast?**

Rerouting options and capacity constraints

The principal bypass routes are:

- **Saudi Arabia East-West (Petroline/Abqaiq–Yanbu)**: design capacity of **~5 mb/d**, reported capacity increases to **7 mb/d** have been claimed but are not fully tested sustainably; IEA estimates **~3–5 mb/d spare** depending on conditions and export logistics on the west coast. ²
- **UAE Abu Dhabi Crude Oil Pipeline (Habshan–Fujairah)**: reported capacity close to **1.8 mb/d**, with roughly **~1.1 mb/d used** and therefore up to **~0.7 mb/d additional** possible in a closure. ²
- **Effective total bypass capacity**: IEA summarizes **~3.5–5.5 mb/d** available to export crude on alternative routes (Saudi + UAE). ²
- **Conservative “effective unused” bypass**: EIA estimates about **2.6 mb/d** could be available (noting day-to-day utilization limits spare). ³

For LNG, constraints are tighter: the IEA emphasizes that Qatari and UAE LNG exports **transit Hormuz almost entirely**, and that there are **no alternative routes** that can bring these volumes to the global LNG market at comparable scale. ⁵

Scenario outcomes

Assumptions for the quantitative shortfall estimates below: - Baseline “at-risk” oil export volume = **19.87 mb/d** (IEA 2025 exports via Hormuz). ²⁰

- Bypass capacity range = **2.6–5.5 mb/d**, with a central case **4.5 mb/d** (IEA range; EIA conservative). ²¹

- Bypass ramp-up occurs over **~2 weeks** (logistics and port/pipeline scheduling), after which the shortfall stabilizes at baseline minus bypass. This is a modeling assumption to reflect operational mobilization, not a sourced “fact.”

Estimated immediate oil supply shortfall (mb/d): in a full closure, the stabilizing net shortfall is approximately: - **High bypass (5.5 mb/d): ~14.4 mb/d** - **Central (4.5 mb/d): ~15.4 mb/d** - **Low bypass (2.6 mb/d): ~17.3 mb/d**

All derived from IEA baseline export volume and IEA/EIA bypass capacity constraints. ⁴

Estimated LNG supply shock (order of magnitude): closure would strand most Qatari and UAE LNG that traverses Hormuz, which the IEA puts at **just over 112 bcm in 2025 (~20% of global LNG trade)** and describes as a drop of **>300 mcm/d** in global LNG supply. ²²

Comparative scenario table

Scenario	Duration	Net oil export loss after rerouting (mb/d)	LNG disruption	Time to restore flows (qualitative)	Key binding constraints
A	1 month	~15.7–17.9 avg over month; stabilizes ~14.4–17.3	Severe; LNG largely stranded	Potentially rapid if military/political de-escalation; market still prices risk	Limited bypass; refined-product exports hardest to reroute; LNG lacks alternatives ²³
B	3 months	~14.8–17.5 avg over period; stabilizes ~14.4–17.3	Severe and prolonged	Restoration depends on conflict resolution; longer disruption forces demand destruction	Same as A, plus inventory depletion and refinery/feedstock dislocations ²³

Scenario	Duration	Net oil export loss after rerouting (mb/d)	LNG disruption	Time to restore flows (qualitative)	Key binding constraints
C	6 months	~14.6–17.4 avg over period; stabilizes ~14.4–17.3	Severe and long enough for power/industry curtailment	Partial normalization may be slow even after reopening due to risk premiums	Multi-month LNG scarcity; macro feedback via inflation/financial tightening ²⁴
D	Indefinite (until market rebalances)	~14.4–17.3 persists until major demand destruction/new supply	Structural LNG deficit relative to global trade	Full rebalancing could take quarters/years	Bypass capacity ceiling; Saudi spare production less useful if export constrained; LNG has no route substitutes ²⁵

Plausible oil and gas price trajectories using elasticities and historical analogs

Anchors from historical oil shocks

Oil price behavior in major disruptions is shaped not only by physical supply losses but also by expectations, precautionary demand, and inventories. Still, historical episodes demonstrate the order of magnitude of price responses to multi-mb/d disruptions:

- **1973–74 oil shock:** production cuts/embargo “nearly quadrupled” the price of oil, from **\$2.90/bbl** to **\$11.65/bbl** by January 1974 (Federal Reserve History). ²⁶
- **1978–79 oil shock:** Iranian output decline of **4.8 mb/d** ($\approx 7\%$ of world production at the time) was associated with oil prices **more than doubling** between April 1979 and April 1980 (Federal Reserve History). ²⁷
- **1990 Gulf crisis:** Brent monthly averages show a rise from **\$17.17/bbl (Jul 1990)** to **\$34.90/bbl (Sep 1990)** and **\$36.02/bbl (Oct 1990)**—approximately a doubling—before easing (EIA Brent historical series). ²⁸

A Hormuz closure removing ~ 15 mb/d net from global seaborne supply (after rerouting) is **multiple times larger** than the physical production losses in these historical oil shocks, implying that **price spikes could be at least comparable and plausibly larger**, especially if LNG scarcity simultaneously raises global fuel-switching demand for oil products.

Illustrative oil price paths by scenario

The chart below is an **illustrative stress path** (not a forecast) showing how prices might evolve under different durations of “complete closure,” using (i) the scale of the net physical shortfall described above and (ii) historical precedent for disproportionate price responses under inelastic demand.

```
xychart-beta
  title "Illustrative Brent price paths under Hormuz closure (index,
baseline=100)"
  x-axis ["M0", "M1", "M2", "M3", "M4", "M5", "M6", "M7", "M8", "M9", "M10", "M11", "M12"]
  y-axis "Price index" 50 --> 300
  line "A: 1 month" [100,230,150,120,115,110,108,106,104,103,102,101,100]
  line "B: 3 months" [100,250,220,200,150,140,130,125,120,115,112,110,108]
  line "C: 6 months" [100,270,260,250,240,230,220,170,160,150,140,135,130]
  line "D: prolonged" [100,280,275,270,265,260,255,250,245,240,235,230,225]
```

Why these magnitudes are plausible: Published IMF research cited by the IMF’s Chief Economist implies even a **10% oil price rise** is macro-meaningful (–0.15% global output next year, +0.4 pp global inflation). ²⁹

A sustained multi-month closure that forces prices to remain substantially higher would therefore create global-scale macro headwinds. In parallel, the IEA stresses that LNG losses of almost 20% of global LNG trade cannot be replaced quickly, which would amplify energy inflation. ²²

Macroeconomic impacts and recession probability by scenario

How oil and gas shocks transmit to recession risk

A Hormuz closure is not just a “price shock”; it is also a **terms-of-trade and real income transfer** from importers to exporters (though exporters may be unable to sell, depending on closure severity). The Euro area’s 2021–2022 energy terms-of-trade shock, for example, was associated with large losses in aggregate income (ECB analysis notes **~1.7 percentage points of GDP** reduction in aggregate income between Q3 2021 and Q3 2022 linked to the energy import shock). ³⁰

Inflation effects can be immediate. A Federal Reserve DSGE analysis finds oil price shocks sized to match the first half of 2022 “noticeably affected headline inflation,” adding **almost 1 percentage point upon impact**, while effects on core inflation and GDP were “fairly small” in that model. ³¹ In a Hormuz closure, the shock is potentially larger and broader (oil + LNG + shipping insurance/freight), increasing the risk that second-round inflation dynamics and tighter monetary policy become recessionary.

A “global recession” benchmark used by the World Bank is a year in which **global real GDP per capita contracts**, with broad weakness across indicators; it identifies global recessions in **1975, 1982, 1991, and 2009** on this basis. ³²

Trade-balance exposure for key importers

Using IEA’s 2025 destination volumes for crude + oil products routed via Hormuz, the incremental annual import bill is roughly **\$19.7B for China per +\$10/bbl**, **\$10.2B for India**, and **\$30.3B for “Other Asia”** (which

includes Japan, South Korea, and other Asian importers). ¹⁷ These arithmetic exposures help explain why Asia bears the most direct physical risk from a closure.

Illustrative import-bill sensitivity (based only on IEA Hormuz-routed volumes, not total oil imports):

Destination (IEA 2025)	Hormuz-routed crude+products (mb/d)	Import bill increase per +\$10/bbl (US\$ bn/year)
Other Asia	8.3	30.3
China	5.4	19.7
India	2.8	10.2
Europe	1.1	4.0
Africa	0.8	2.9
Americas	0.7	2.6

Computed from IEA destination volumes. ¹⁷

Scenario-by-scenario macro impacts

The table below provides approximate *directional* macro shocks, anchored to IMF's oil price multiplier and adjusted qualitatively for LNG-specific stresses (especially in South Asia and parts of Asia with high Qatari/UAE LNG reliance). ⁶

Scenario	Expected global inflation impulse	Expected global GDP impulse	Most vulnerable major economies	Global recession probability (structured judgment)
A: 1 month	Noticeable but likely partly transitory; strong headline CPI jump	Growth hit likely measurable but often not enough alone to force global per-capita contraction	Asia most directly (oil + LNG), but global pricing transmits everywhere ³³	~15–30%
B: 3 months	Higher risk of second-round effects and tighter financial conditions	Larger hit; demand destruction begins; industrial curtailments possible in gas-short importers	China/India + “Other Asia” (Japan/Korea), plus LNG-dependent emerging Asia/South Asia ³⁴	~35–55%

Scenario	Expected global inflation impulse	Expected global GDP impulse	Most vulnerable major economies	Global recession probability (structured judgment)
C: 6 months	Persistent energy inflation; higher odds monetary tightening stays restrictive	Broad output drag; some regions likely enter recession even if global aggregate stays positive	Asia widely; Europe via LNG competition and price transmission; EM importers with weak FX buffers ³⁵	~55–75%
D: prolonged	Energy inflation regime shift; material risk to inflation expectations in some jurisdictions	High likelihood of global per-capita GDP contraction; widespread rationing/curtailment in weakest importers	Asia + energy-intensive EMs; global spillovers through trade, finance, and confidence ³⁶	~75–90%

Key country/region notes tied to the data: - **China and India** are the single largest identified crude destinations in IEA data (4.6 and 2.1 mb/d of crude via Hormuz in 2025, respectively). ¹⁷

- **Japan and Korea** are highlighted as particularly reliant among IEA countries, and EIA emphasizes that the top Asian importers (China, India, Japan, South Korea) collectively account for ~69% of Hormuz crude/condensate flows (2024). ³⁷

- **Europe's direct physical volume via Hormuz is smaller** (~0.6 mb/d crude and ~0.5 mb/d products in IEA's 2025 routing depiction), but Europe remains vulnerable via global price transmission and LNG tightness. ³⁴

- **South Asia and LNG-dependent emerging markets** face outsized risks: IEA notes Bangladesh, India, and Pakistan imported almost **two-thirds of their total LNG supplies via Hormuz** in 2025, implying potential electricity and industrial disruption under LNG scarcity. ³⁸

Sensitivity analysis, assumptions, and data gaps

Key sensitivities that swing recession outcomes

The recession question is extremely sensitive to a small number of parameters:

The effective bypass capacity is pivotal. If bypass is **closer to EIA's ~2.6 mb/d** rather than the upper end of IEA's **3.5–5.5 mb/d** range, the net oil shortfall rises by several mb/d—enough to materially change the required price increase and the duration that inventories can buffer the shock. ³⁹

Duration is nonlinear. A one-month closure can be bridged more plausibly by inventories and emergency measures; beyond three months, the world increasingly must clear through demand destruction, forced substitution, and macro tightening—which increases recession odds.

The LNG channel can dominate in certain regions. IEA's assessment that a Hormuz closure would strand a large share of global LNG trade and cannot be replaced quickly implies that gas markets could experience acute scarcity, potentially forcing power rationing and industrial curtailments in the most exposed economies. ²²

Data gaps and measurement caveats

Ship transit counts are subject to methodological caveats. ONS describes these statistics as “official statistics in development” and notes AIS data issues (noise, gaps, tampering, and missing ship-type information). ⁴⁰ The ship-type counts used here (oil and liquefied gas) are therefore best treated as **high-quality indicators**, not perfectly accurate censuses.

“Liquefied gas” transits are not identical to “LNG” transits. ONS ship types aggregate liquefied gas and other liquids; LNG volumes are better anchored by EIA and IEA flow estimates (e.g., EIA's ~10–11+ Bcf/d and IEA's ~112 bcm in 2025). ⁴¹

Tonnage measures differ from ship counts. UNCTAD's gross-ton transits measure captures the “mass” of vessel traffic via Clarksons, not the number of vessels; both are useful but not interchangeable. ¹⁶

Price paths are scenario constructs. The price trajectories shown are not sourced “forecasts”; they are stress paths designed to be consistent with the scale of physical shortfalls (IEA/EIA) and with the historical tendency of oil prices to overshoot under large, sudden supply risks (1973–74 and 1978–79 Federal Reserve History; 1990 Brent series from EIA). ⁴²

Bottom line

A **short closure** of Hormuz is not guaranteed to cause a worldwide recession, but it would likely produce a sharp global inflation shock and serious regional downturn risks. A **multi-month closure** has a high likelihood of triggering recession in multiple major economies and a substantial probability of a global recession as defined by global per-capita GDP contraction—because the at-risk oil volumes are ~20 mb/d with only limited bypass options, and LNG losses are close to one-fifth of global LNG trade with essentially no route substitutes. ⁴³

¹ ² ⁴ ⁵ ⁸ ⁹ ¹⁰ ¹⁷ ¹⁸ ²⁰ ²¹ ²² ²³ ²⁴ ²⁵ ³⁴ ³⁷ ³⁸ ⁴² ⁴³ <https://iea.blob.core.windows.net/assets/c8248eba-8689-46d9-ae4b-b858b59c0f1c/StraitofHormuz2026-Factsheet.pdf>

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³ ¹² ¹⁹ ³³ ³⁹ <https://www.eia.gov/todayinenergy/detail.php?id=65504>

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⁶ ⁷ ²⁹ <https://mediacenter.imf.org/news/imf-world-economic-outlook-presser/s/e7c7583c-9bae-45f5-a44e-377e7082554e>

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